ELAN

SANTÉ

ISIS Ingénieurs - Promo 2027

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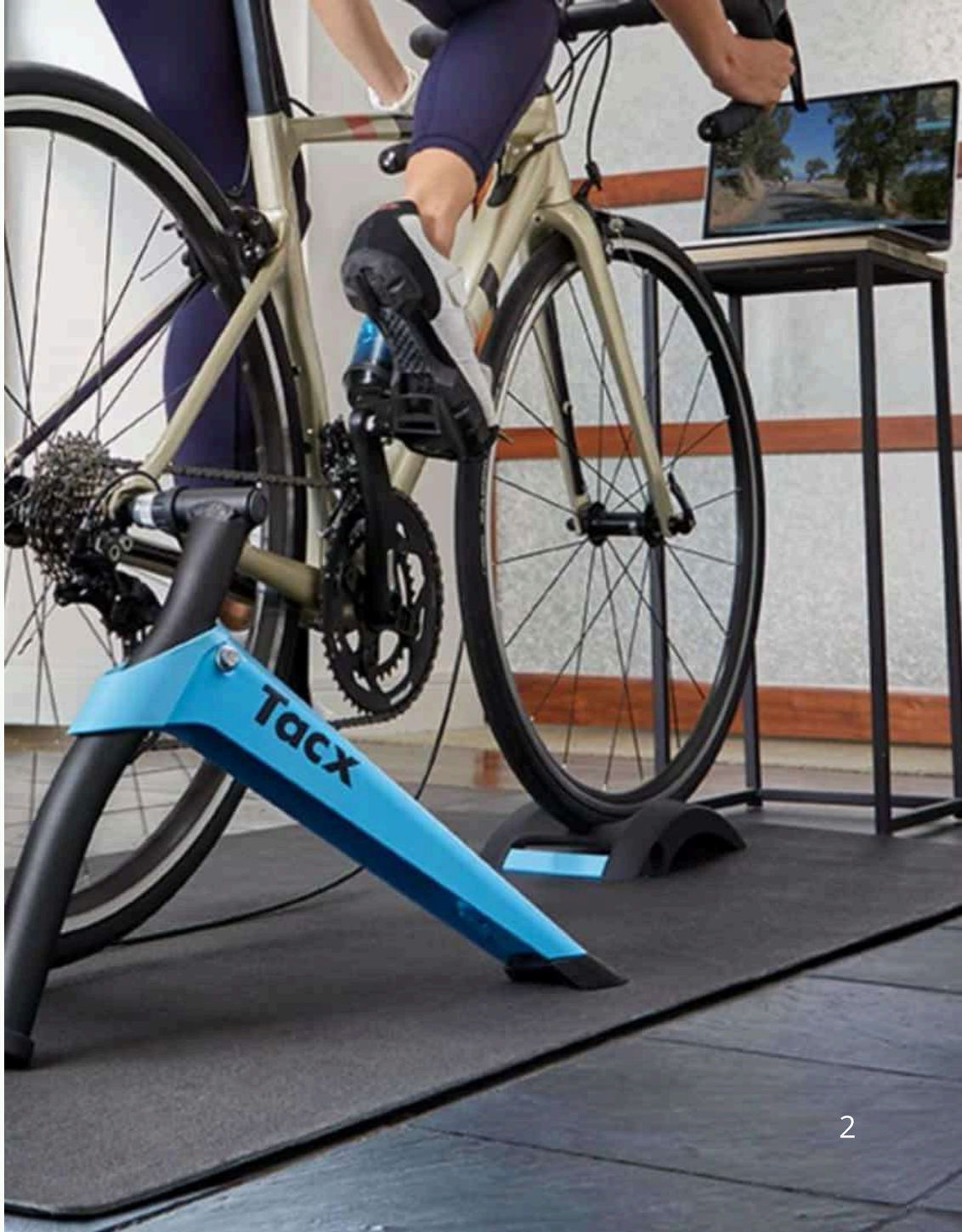
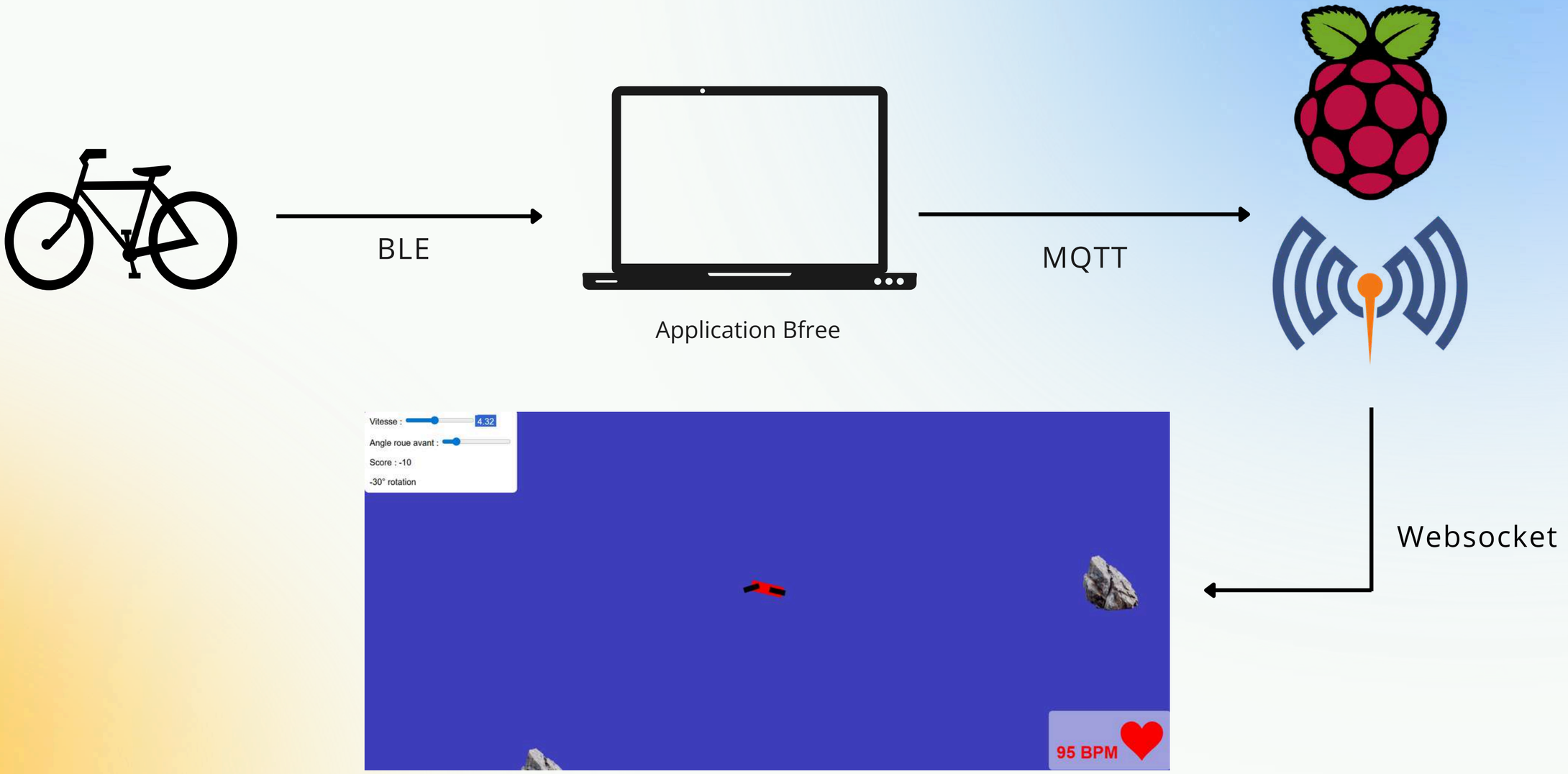


INTRODUCTION

01 Un projet IoT au service de la rééducation physique

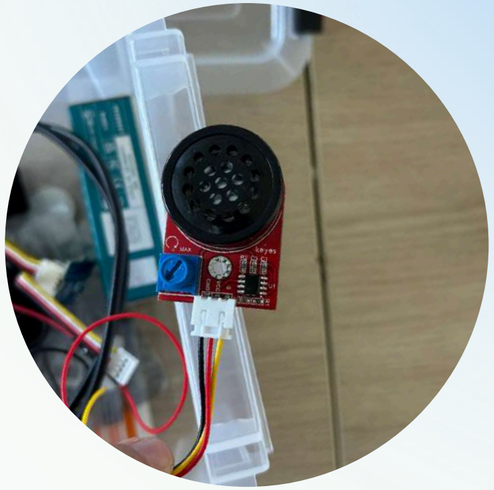
02 Le relais connecté – duo actif et ludique !

1. Dimension technique : la vitesse



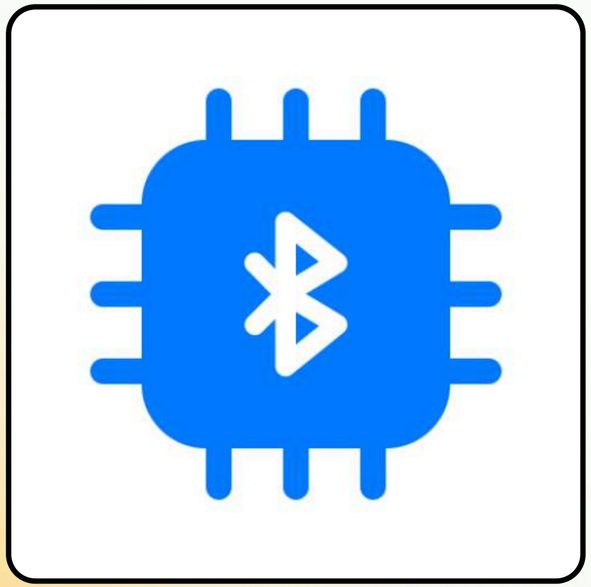
1. Dimension technique : autres composants du système

Un capteur de pouls

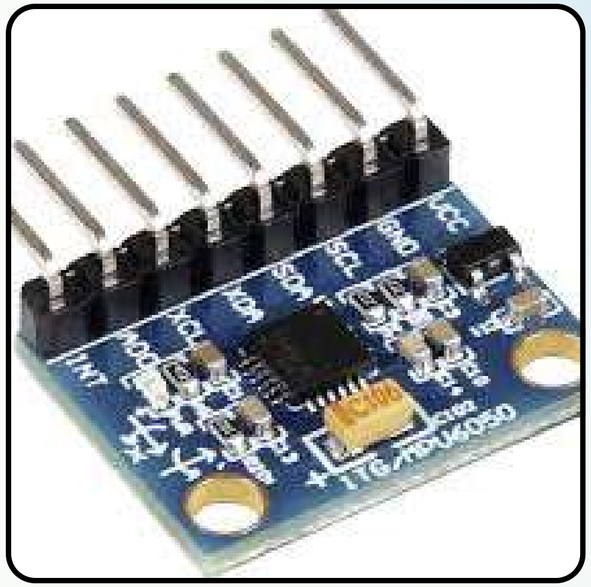


Buzzer

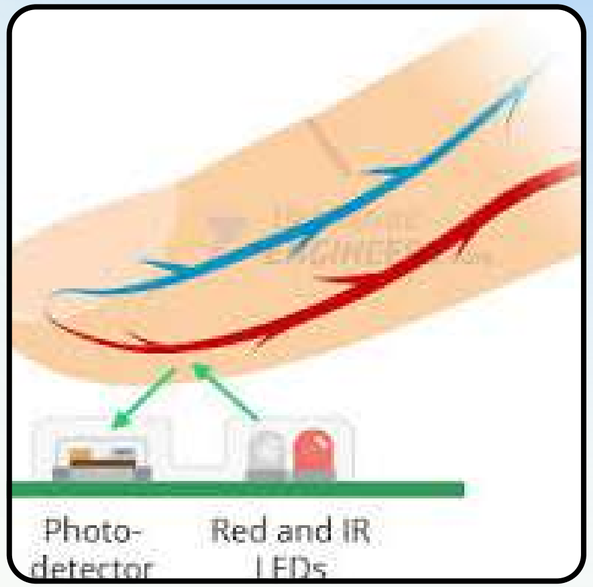
2. Dimension technique



Contrôle via Blue Remote (manuel)
Angle de direction saisi dans l'app

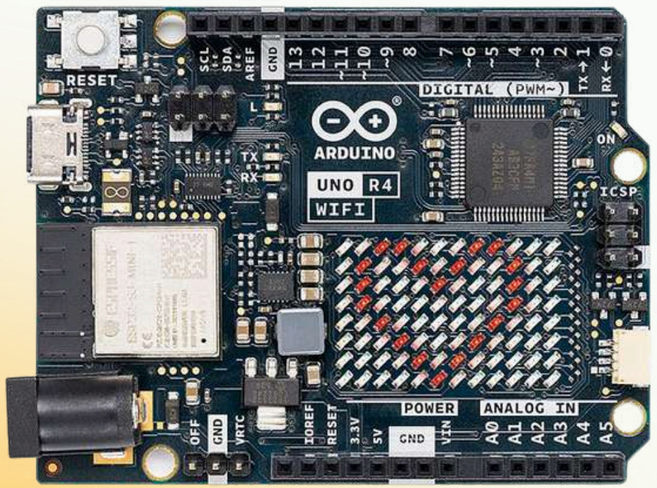


Contrôle manuel → capteur physique
Angle via accéléromètre (MQTT)



Constantes vitales en temps réel
FC (BPM) · SpO2 · Température

2. Dimension technique



HomeTrainer/angle

HomeTrainer/SPO2

HomeTrainer/BPM

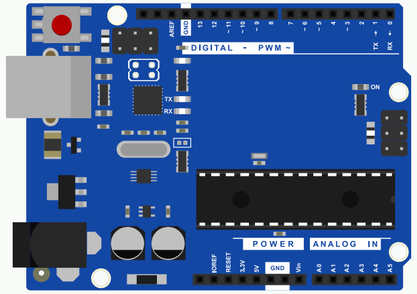
HomeTrainer/Temp

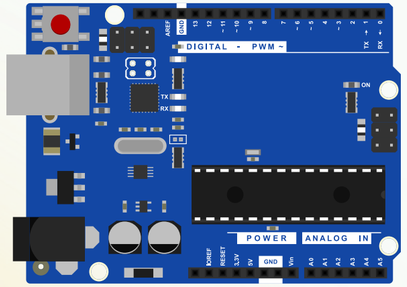
MQTT Broker : Mosquitto

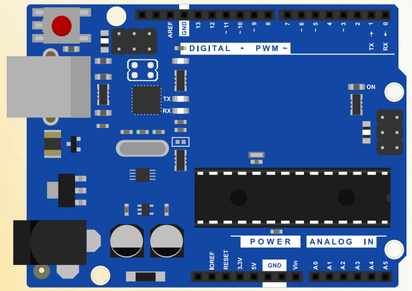
192.168.0.227

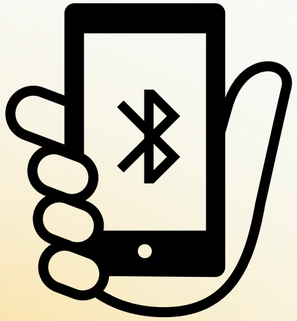
Le relais en duo

Publishers







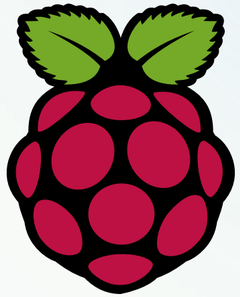


Beacon

isisChallenge/borne/B012

isisChallenge/borne/101

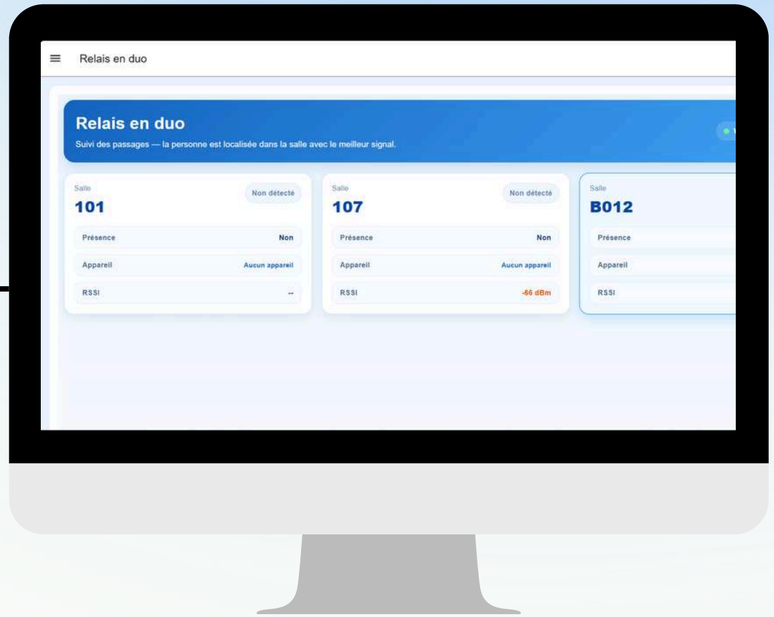
isisChallenge/borne/107



MQTT Broker : Mosquitto

192.168.0.227

Subscribers

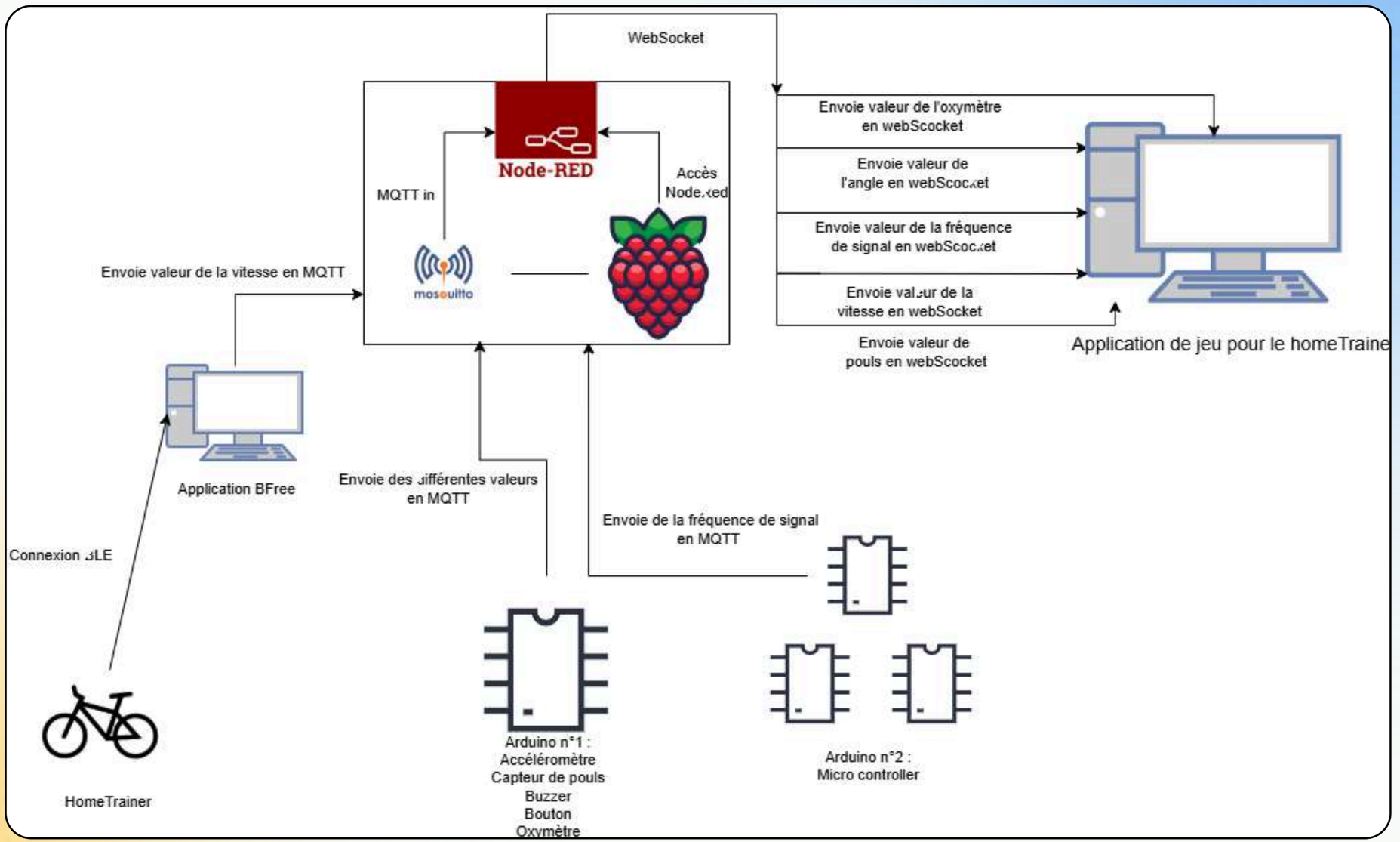


→ RSSI mesuré par chaque borne

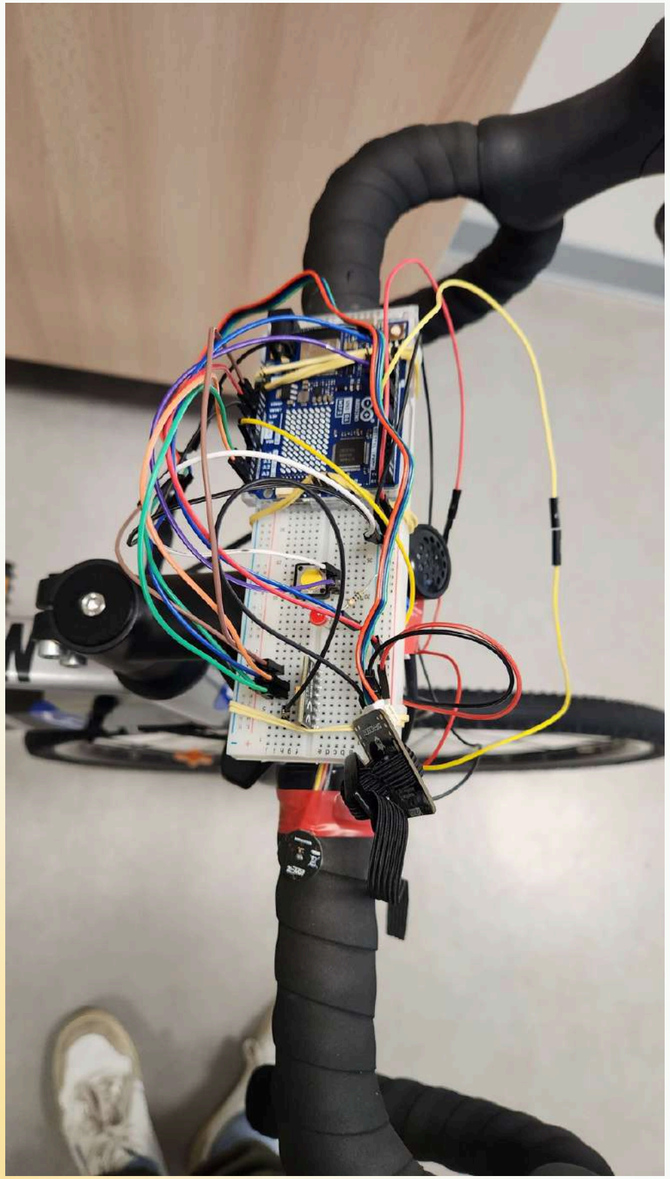
→ Payload : valeur RSSI (-46, -100...)

→ WebSocket port 1880

Architecture



Montage



Un capteur de pouls

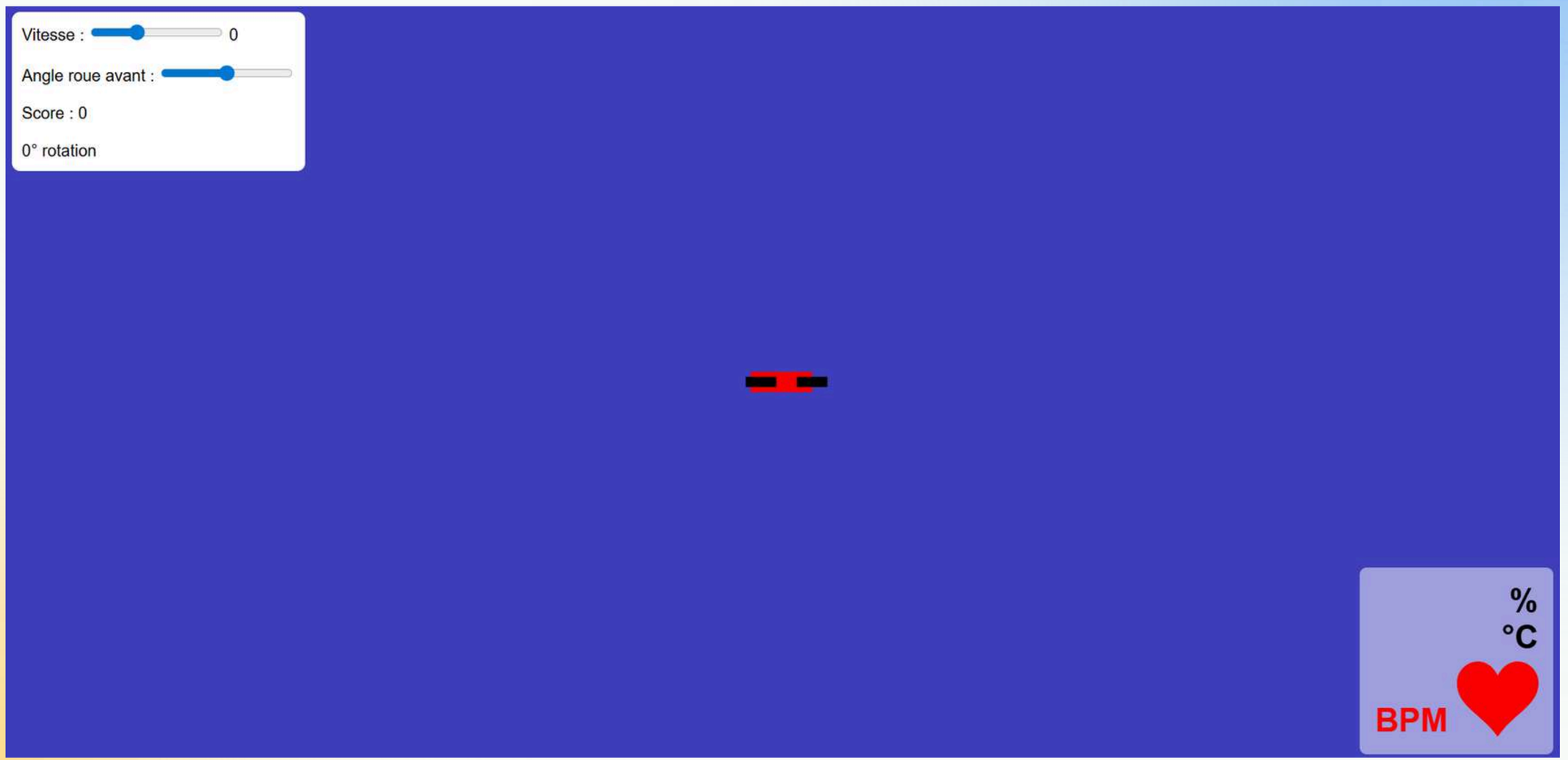
Un oxymètre

Un bouton

Un Buzzer

Une Led

Le jeu rééducatif



DÉMONSTRATION

Les limites

01 Capteurs sensibles / calibration complexe

02 Absence de validation médicale

03 Sécurité des données non traitée

Les perspectives

01 Un suivi personnalisé

02 Adaptation au patient

03 Collaboration avec des professionnels de santé

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